Fiscal Year:	FY 2005	**Task Last Updated:**	FY 04/25/2005
PI Name:	Lewis, Jennifer A. Sc.D.		
Project Title:	Directed Assembly of Hydroxyapatite Scaffolds		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Other		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jalewis@uiuc.edu	**Fax:**	FY 217 244-6917
PI Organization Type:	UNIVERSITY	**Phone:**	217 244-4973
Organization Name:	University of Illinois-Urbana Champaign		
PI Address 1:	Materials Science and Engineering Department		
PI Address 2:	1304 W. Green Street		
PI Web Page:	http://colloids.mse.uiuc.edu/ ; http://www.mse.uiuc.edu/		
City:	Urbana	**State:**	IL
Zip Code:	61801	**Congressional District:**	15
Comments:			
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	00-HEDS-UNSOL
Start Date:	07/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Gilles, Donald	**Contact Phone:**	256 544-9302
Contact Email:	donald.C.Gillies@msfc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Jamison, Russ (University of Illinois)		
Grant/Contract No.:	NAG8-1922		
Performance Goal No.:			
Performance Goal Text:			

PI Lewis and coPI Jamison have created 3-D periodic hydroxyapatite (HA) scaffolds with controlled multiscale porosity. Concentrated colloidal HA inks have been formulated with tailored viscoelastic properties required for direct-write assembly. 3-D scaffolds have been fabricated via robotic deposition of these inks in a layer-by-layer build sequence. PI Lewis has fully characterized with respect to their pore structure at the nanoscale (< 100 nm), microscale (100 nm to 100 microns), and mesoscale (> 100 microns) using both direct imaging techniques and indirect structural probes (e.g., mercury intrusion porosimety). co-PI Jamison has produced scaffolds of several designs in largequantities (> 100 scaffolds) to enable characterization and optimization of various structural attributes of interest. We focused specifically in year 2 on understanding the influence of interconnected microposity (~6 µm diameter) on initial mechanical properties of the scaffolds, degradation of the scaffolds in simulated biological conditions, and interaction of bone cells (osteoblasts) with these micropores in culture conditions. We established methods to reproducibly create this interconnected microporosity in HA scaffolds by loading the HA gel with fugitive polymethylmethacrylate (PMMA)

Task Description:	beads during the directed deposition step of scaffold fabrication. The broad objectives of this ground-based research program are to fabricate and characterize mesoporous hydroxyapatite (HA) structures with microporous domains that serve as model tissue engineering scaffolds. Specifically, model HA scaffolds are produced via a directed colloidal assembly technique known as robotic deposition. We then characterize their cellular response through in vitro studies. Our ultimate goal is to develop a fundamental understanding of the interactions between bone tissue and HA scaffolds, and, from that understanding, establish guidelines for the rational design of hydroxyapatite scaffolds that optimize the rate and extent of tissue ingrowth. We expect the knowledge and experience gained in producing diverse architectures of hydroxyapatite to provide valuable insights on producing similar scaffolds from polymers or polymer/HA composites. Such scaffolds could be used not only for bone repair, but for regeneration of skin and other tissues in the body. The versatility, simplicity, and compactness of the robotic deposition equipment used in this research would therefore meet NASA's long-term goal of creating a multifunctional materials platform that can be used as a controllable scaffold for the guided regeneration of a broad spectrum of tissue systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The ability to produce engineered, 3-D scaffolds for bone generation could have a profound impact on surgical procedures involving bone grafts, which are currently highly invasive.
Task Progress:	We have produced scaffolds of several designs in large quantities (> 100 scaffolds) to enable characterization and optimization of various structural attributes of interest. We focused specifically in year 2 on understanding the influence of interconnected microposity (~6 µm diameter) on initial mechanical properties of the scaffolds, degradation of the scaffolds in simulated biological conditions, interaction of bone cells (osteoblasts) with these micropores in culture conditions, and the effects of mesoscale porosity (> 100 microns) on bone ingrowth. Both 3-D periodic and quadrant designs were pursued. We designed and completed a series of in vitro experiments to simulate the degradation behavior of HA scaffolds having either dense or microporous structure. We accelerated the degradation by buffering the medium to pH 4.0 by the addition of sodium acetate following the method of Koerten et. al.[1] Scaffolds were nominally 4.5 mm in height and 11.5 mm in diameter. Scaffolds consisted of an outer wall nominally 1.5 mm thick and an inner rectilinear arrangement of HA rods of 570 µm diameter. Inter-rod spacing, both lateral and vertical was either 270 µm or 680 µm. Samples were weighed, placed in a 0.1M sodium acetate solution buffered to pH 4.0 and kept in a shaking water bath at 37ºC for various periods of time ranging from 12 h to 336 h. The longest time represents approximately one year of in vivo implantation.1 Samples were dried and tested for compression strength and elastic modulus as well as mass loss. The solution was filtered and assay for identification of degradation products. The compression strength and modulus of non-microporous scaffolds measured after 336 h of degradation were not statistically different from initial values. Initial and residual values of strength and stiffness for these scaffolds were 302 ± 55 Mpa and 5.1 ± 0.3 GPa respectively. The same properties for microporous scaffolds showed a monotonic decrease with degradation time from initial values of 10 ± 2 MPa and 0.5 ± 0.2 GPa to 6.0 ± 0.5 MPa and 0.3 ± 0.1 GPa. These values represent reductions of 40% over initial properties. The results of these experiments suggest that fully dense scaffolds exhibit strength and stiffness comparable to human cortical bone after simulated degradation, whereas fully microporous scaffolds do not. These results provide bounds for design of load-bearing scaffolds having a combination of dense and microporous elements. We will be examining scaffolds of the same design retrieved from intramuscular and intercortical implantation in animals. These results will be provided in year 3. HA scaffolds fabricated with rods containing interconnected microporosity presented a significantly different surface morphology than identical scaffolds with rods comprised entirely of dense HA. We investigated the effects of these morphological differences on cellular interactions holding all other attributes of the scaffolds fixed. The length scale of the surface pits created by micropores (~6 µm) was comparable to the size of an individual cell (~10 µm). Surface roughness (Ra) for dense and microporous rods was 0.8 ± 0.2 µm and 1.3 ± 0.2 respectively as determined by contact profilometry. HA scaffolds were cultured with mouse mesenchymal stem cells (D1 ORL UVA) in 10% bovine serum supplemented with ascorbic acid and ß-glycerolphosphate to induce osteoblastic differentiation. (We use adult stem cells in these experiments to assess the potential clinical utility of using a patient's stem cells to seed tissue engineered scaffolds in therapeutic applications.) Differentiated cells were maintained in culture and removed for examination at intervals up to 21 days. At early time points (less than 10 days) there appeared to be morphological differences in the cells attached to dense and microporous HA compared to polystyrene (PS) controls; but by 21 days those differences had disappeared. SEM examination of cells attached to the two types of scaffolds and PS showed direct involvement of cell processes with micropores. Cells attached to the microporous surfaces showed intimate and extensive engagement with microporosity. It was therefore surprising that measurements of cell number and alkaline phosphatase (ALP) expression (a phenotypic marker of osteoblasts activity) on these surfaces were lower than controls. ALP activity of cells was delayed and extracellular matrix expression was more substantial in these cases. This suggests that the surface morphology affects cells in unexpected ways. The attachment of osteoblasts to HA surfaces generally and microporous HA surfaces specifically is suggestive of strong binding. In experiments in which cells are removed from surfaces using trypsin, the cells attached to microporous surfaces are substantially more adherent. Our current work in this area involves characterization of cell attachment to scaffolds from animal implantation for comparison with these in vitro results. One rationale for designing and characterizing HA scaffolds with microporous rods is the opportunity they present for delivery of growth factors to aid in the regeneration of tissue within and adjacent to the scaffold. The interconnected nature of this microporosity provides a relatively large volume and surface area to contain drugs. We have completed a series of in vitro experiments to characterize the uptake and release of a model drug in order to assess the feasibility of such a feature. We chose bovine serum albumin (BSA) as a model drug because of its relatively lower cost and similar molecular weight to bone morphogenetic protein 2 (BMP-2), a potent bone growth factor. Scaffolds were immersed in a solution of BSA and phosphate buffered saline (PBS) and allowed to absorb the solution for ten minutes at room temperature. Scaffolds were removed, submerged in vials of fresh PBS and gently agitated at 37ºC for periods up to 168 hours. Some drug-loaded scaffolds were lyophilized before re-immersion to assess the potential to store drug-bearing scaffolds prior to reconstitution as a clinical approach. At intervals, the concentration of BSA in solution was measured using a micro-bicinchoninic acid assay and spectrophotometry. Elution profiles of BSA from lyophilized and non-lyophilized microporous scaffolds displayed logarithmic behavior, suggesting a diffusion-dominated process as expected. At seven days the drug was completely released from both

	sample types with no statistically significant difference in release rates. These results suggest that lyophilization of drug-bearing scaffolds may be a possible route to cost-effective clinical application. The seven day release period may be sufficient for therapeutic action of growth factors like BMP-2 which are know to be present in bone healing in the first 3-4 days. Our current work in this area is focused on measuring the release of BMP-2 and determining its viability after elution and lyophilization. We have recently implanted microporous HA scaffolds containing BMP-2 into the metatarsal bone of goats to assess in vivo mechanisms and response. Those results will be reported in year 3. We have also recently implanted HA scaffolds with a quadrant architecture into white New Zealand rabits, to assess in vivo bone ingrowth and make direct comparisons of their response to HA scaffolds created by other direct writing approaches, such as 3D printing. These results will also be reported in Year 3.
Bibliography Type:	Description: (Last Updated: 12/21/2007)
Articles in Peer-reviewed Journals	Michna S, Wu W, Lewis JA. "Concentrated Hydroxapatite Inks for Direct-Write Assembly of 3-D Periodic Scaffolds." Biomaterials, in press, 2005. , Jul-2005
Dissertations and Theses	Delllinger J. "Development of Model Hydroxyapatite Bone Scaffolds with Multiscale Porosity for Potential Load Bearing Applications." Ph.D. Dissertation, University of Illinois at Urbana-Champaign, February, 2005. , Feb-2005
Dissertations and Theses	Michna S. "Concentrated Hydroxapatite Inks for Direct-Write Assembly of 3-D Periodic Scaffolds." M.S. Thesis, University of Illinois at Urbana-Champaign, December, 2004. , Dec-2004